

ISO15693 Air Interface Protocol Function Library
v1.4

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1. Introduction

The development kit supports the ANSI-C, ANSI-C++ and other languages to call the C function interface, such as delphi, VB, C#. The development kit can not be used independently and must be combined with other RFID development kit. Before using, you should read "DLL Development Guide.doc" first, to make sure well understand the whole development process for SDK package.

2. Application Integration

The SDK consists of the following files:

Documents	Application
rfidlib_aip_iso15693.dll	DLL includes all API interface
rfidlib_aip_iso15693.lib	VC++ project static link library
rfidlib_aip_iso15693.h	C/C++ project header file, function definition
rfidlib_aip_iso15693.cs	C# function definition file
rfidlib_aip_iso15693.pas	Delphi function definition header file

Programming Language	Instruction
VC++	Copy rfidlib_aip_iso15693.lib and rfidlib_aip_iso15693.h to the project directory and add to the project. Copy rfidlib_aip_iso15693.dll to exe file output directory.
C#	Copy rfidlib_aip_iso15693.cs to the project directory. Copy rfidlib_aip_iso15693.dll to exe file output directory.
Delphi	Copy rfidlib_aip_iso15693.pas to the project directory. Copy rfidlib_aip_iso15693.dll to exe file output directory.

3. Programming Interface

3.1. Overview

The programming interface is based on rfidlib_reader.dll, definite ISO15693 standard protocol command interface and convenient application programming developer use ISO15693 protocol.

3.2. ISO15693_GetLibVersion 【Get Library Version】

Definition	DWORD ISO15693_GetLibVersion(LPTSTR buf , DWORD nSize)		
Instruction	Get the version of dynamic library		
Parameter	Name	Type	Remark
	buf	LPTSTR	String buffer, used for saving the version number.
	nSize	DWORD	The maximum savable string in buffer
Return (DWORD)	Returns the number of characters written to the buffer, if the return value is equal to "nSize", shows that the version number length is bigger than buffer capacity.		
Reference Code	<pre>char vbuf[128] ; DWORD written ; written = get_version(vbuf,128) ; if(written > 0) { /* show version string here */ printf(vbuf) ; } if(written == 128) { /* overflow,need to increase vbuf */ }</pre>		

3.3. ISO15693_CreateInvenParam 【Create Inventory Parameter】

Definition	RFID_DN_HANDLE ISO15693_CreateInvenParam(RFID_DN_HANDLE hInvenParamSpecList, BYTE AntennaID, BOOLEAN en_afi, BYTE afi, BYTE slot_type)				
Instruction	Create ISO15693 Inventory parameter data node, used in the API of “RDR_TagInventory”				
Parameter	Name	Type	Direction	Remark	
	hInvenParamSpecList	RFID_DN_HANDLE	[in]	Inventory parameter list handle, created by RDR_CreateInvenParamSpecList.	
	AntennaID	BYTE	[in]	Antenna ID, 0 means all of the antenna interface will read the tags which supported ISO15693 protocol. Other values for the corresponding antenna interface.	
	en_afi	BOOLEAN	[in]	On the process of ISO15693 Inventory, match AFI value or not. 0: No; 1: yes.	
	Afi	BYTE	[in]	AFI value	
	slot_type	BYTE	[in]	ISO15693 Inventory Slot Types	
				ValueInstruction	
				0	Default slot
				1	1 slot
			16	16 slot	
Return (RFID_DN_HANDLE)	Success: return the handle of data node.				
Reference Code					

3.4. ISO15693_ParseTagDataReport 【Parse Inventory Report】

Definition	err_t ISO15693_ParseTagDataReport(RFID_DN_HANDLE hTagReport, DWORD *aip_id,DWORD *tag_id,DWORD * ant_id, BYTE *dsfid,BYTE uid[])			
Instruction	Parse ISO15693 Inventory tag data information. Obtained data node from RDR_GetTagDataReport, for the RDR_TagInventory function.			
Parameter	Name	Type	Direction	Remark
	hTagReport	RFID_DN_HANDLE	[in]	The node handle of the tag data information
	aip_id	DWORD *	[out]	Returns to the air interface protocol type ID, please refer to the air interface protocol type ID list of “DLL Development Guide.doc”.
	tag_id	DWORD *	[out]	Returns to the tag types ID , refer to the tag types ID list of “DLL Development Guide.doc”.
	ant_id	DWORD *	[out]	Return antenna interface ID
	dsfid	BYTE *	[out]	Return DSFID of tag message for ISO15693
	uid	BYTE []	[out]	Return serial number of tag message for ISO15693
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.5. ISO15693_TransparentTransceive 【TransparentTransceive】

Definition	err_t RFIDLIB_API ISO15693_TransparentTransceive(RFID_READER_HANDLE hr, BYTE mode, BYTE IsTxCRC, BYTE IsRxCRC, BYTE *reqData, DWORD reqLen, DWORD nBytesTagWillRet, BYTE *pRxBuffer, DWORD *nSize, DWORD time)			
Instruction	ISO15693 protocol transparent transceive command			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader handle
	mode	BYTE	[in]	Mode 0x00 : read command 0x01 : write command with option = 0 0x02 : write command with option = 1
	IsTxCRC	BYTE	[in]	Whether auto send CRC
	IsRxCRC	BYTE	[in]	Whether auto receive CRC
	reqData	BYTE *	[in]	The request data which will be send to the tag.
	reqLen	DWORD	[in]	The length of the request data.
	nBytesTagWillRet	DWORD	[in]	The byte count of data for tag return.
	pRxBuffer	BYTE *	[out]	Data buffer for receive data from tag.
	nSize	DWORD *	[in] [out]	[in]The size of pRxBuffer. [out]The size of the data which have been written into pRxBuffer.
	time	DWORD	[in]	When mode = 0x00, parameter retains, set 0; When mode = 0x01, parameter means "time for parse SOF,unit:us" ; When mode = 0x02, parameter means "wait time befor send EOF,unit:us"
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.6. ISO15693_Connect 【Connect Tag】

Definition	err_t ISO15693_Connect(RFID_READER_HANDLE hr , DWORD tagType, BYTE address_mode, BYTE uid[] , RFID_TAG_HANDLE *ht) ;				
Instruction	Connect ISO15693 tag and create corresponding instance handle				
Parameter	Name	Type	Direction	Remark	
	hr	RFID_READER_HANDLE	[in]	Reader handle	
	tagType	DWORD	[in]	Tag type ID , refer to the tag types ID list of “DLL Development Guide.doc”. If set 0, use default tag type of reader	
	address_mode	BYTE	[in]	Addressing mode	
				Value	Instruction
				0	Not Address, not in need of uid.
				1	Address mode, in need of uid.
	2	Select mode , in need of uid.			
	uid	BYTE[]	[in]	Tag serial number, effected when address_mode set 1 or 2	
ht	RFID_TAG_HANDLE *	[out]	Connect success, return tag handle		
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.				
Reference Code					

3.7. ISO15693 Standard Command

3.7.1. ISO15693_ReadMultiBlocks 【Read Multiple Blocks】

Definition	err_t ISO15693_ReadMultiBlocks(RFID_READER_HANDLE hr , RFID_TAG_HANDLE ht, BOOLEAN readSecSta , DWORD blkAddr, DWORD numOfBlksToRead, DWORD *numOfBlksRead, BYTE bufBlocks[], DWORD nSize, DWORD *bytesBlkDatRead) ;			
Instruction	ISO15693 optional command, reader multiple blocks			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle, created by ISO15693_connect
	readSecSta	BOOLEAN	[in]	Whether read data block security status
				Value Instruction
				1 Read block and security byte, such as ICODE SLI; Security byte (1byte) +block(4bytes)=5
				0 Don't read security status, such as ICODE SLI, only have 4 bytes block
	blkAddr	DWORD	[in]	Block address.
	numOfBlksToRead	DWORD	[in]	Number of blocks to read
	numOfBlksRead	DWORD *	[out]	The number of Blocks which have been read.
	bufBlocks	BYTE []	[out]	The Buffer which is used for saving the data.
	nSize	DWORD	[in]	The length of the buffer which is used of saving the data.
	bytesBlkDatRead	DWORD *	[out]	The length of the buffer which have been write successfully.
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.7.2. ISO15693_WriteMultipleBlocks 【Write Multiple Blocks】

Definition	err_t ISO15693_WriteMultipleBlocks(RFID_READER_HANDLE hr , RFID_TAG_HANDLE ht, DWORD blkAddr, DWORD numOfBlks, BYTE newBlksData[], DWORD bytesToWrite) ;			
Instruction	ISO15693 optional command, write multiple blocks			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle, created by ISO15693connect
	blkAddr	DWORD	[in]	Block address.
	numOfBlks	DWORD	[in]	Number of data blocks to write
	newBlksData	BYTE []	[in]	New block data
	bytesToWrite	DWORD	[in]	The number of bytes to write.
Return	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code	No			

3.7.3. ISO15693_LockMultipleBlocks 【Lock Multiple Blocks】

Definition	err_t RFIDLIB_API ISO15693_LockMultipleBlocks(RFID_READER_HANDLE hr , RFID_TAG_HANDLE ht, DWORD blkAddr, DWORD numOfBlks)			
Instruction	Lock multiple blocks.			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle, created by ISO15693_connect
	blkAddr	DWORD	[in]	Address of the data block to be locked
	numOfBlks	DWORD	[in]	The number of block need to be blocked
Return	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.7.4. ISO15693_WriteAFI 【Write AFI】

Definition	err_t ISO15693_WriteAFI(RFID_READER_HANDLE hr , RFID_TAG_HANDLE ht, BYTE afi) ;			
Instruction	ISO15693 optional command, write AFI			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle, created by ISO15693_connect
	afi	BYTE	[in]	AFI Value
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.7.5. ISO15693_LockAFI 【Block AFI】

Definition	err_t ISO15693_LockAFI(RFID_READER_HANDLE hr , RFID_TAG_HANDLE ht) ;			
Instruction	ISO15693 optional command, lock AFI			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle created by ISO15693_connect
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.7.6. ISO15693_WriteDSFID 【Write DSFID】

Definition	err_t ISO15693_WriteDSFID(RFID_READER_HANDLE hr , RFID_TAG_HANDLE ht, BYTE dsfid) ;			
Instruction	ISO15693 optional command, write DSFID			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle, created by ISO15693_connect
	dsfid	BYTE	[in]	DSFID Value
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.7.7. ISO15693_LockDSFID 【Lock DSFID】

Definition	err_t ISO15693_LockDSFID(RFID_READER_HANDLE hr , RFID_TAG_HANDLE ht) ;			
Instruction	ISO15693 optional command, lock DSFID			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle, created by ISO15693_connect
Return err_t	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code	No			

3.7.8. ISO15693_GetSystemInfo 【Get Tag System Information】

Definition	err_t ISO15693_GetSystemInfo(RFID_READER_HANDLE hr , RFID_TAG_HANDLE ht,BYTE uid[], BYTE *dsfid,BYTE *afi, DWORD *blkSize , DWORD *numOfBlocs, BYTE *icRef) ;			
Instruction	ISO15693 optional command, get tag system information			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle, created by ISO15693_connect
	uid	BYTE []	[out]	8bytes buffer, used to save tag series No.
	dsfid	BYTE *	[out]	Used to save tag information DSFID value
	afi	BYTE *	[out]	Used to save tag information AFI value
	blkSize	DWORD *	[out]	Used to save tag information single block length
	numOfBlocs	DWORD *	[out]	Used to save the total number of blocks
	ic_ref	BYTE *	[out]	Used to save tag information IC reference
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.7.9. ISO15693_GetBlockSecStatus 【Get Block Security Status】

Definition	err_t ISO15693_GetBlockSecStatus(RFID_READER_HANDLE hr , RFID_TAG_HANDLE ht, DWORD blkAddr, DWORD numOfBlks, BYTE bufBlkSecs[], DWORD nSize,DWORD *bytesSecRead) ;			
Instruction	Read security status			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle, created by ISO15693_connect
	blkAddr	DWORD	[in]	Block address.
	numOfBlks	DWORD	[in]	Number of Blocks need to be read.
	bufBlkSecs	BYTE []	[out]	The buffer which is used for saving the security status.
	nSize	DWORD	[in]	The buffer size.
Return	0: Success			
	Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.8. NXP ICODE-SLI Self-definition Command

3.8.1. NXPICODESLI_EableEAS 【Open EAS function】

Definition	err_t NXPICODESLI_EableEAS(RFID_READER_HANDLE hr , RFID_TAG_HANDLE ht) ;			
Instruction	Enable EAS function			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle, created by ISO15693_connect
Return	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.8.2. NXPICODESLI_DisableEAS 【Close EAS Function】

Definition	err_t NXPICODESLI_DisableEAS(RFID_READER_HANDLE hr , RFID_TAG_HANDLE ht) ;			
Instruction	Disable EAS function			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle, created by ISO15693_connect
Return	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.8.3. NXPICODESLI_LockEAS 【Lock EAS】

Definition	err_t NXPICODESLI_LockEAS(RFID_READER_HANDLE hr , RFID_TAG_HANDLE ht) ;			
Instruction	Lock EAS function, once locked, unable to unlock			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle, created by ISO15693_connect
Return	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.8.4. NXPICODESLI_EASCheck 【Check EAS】

Definition	err_t NXPICODESLI_EASCheck(RFID_READER_HANDLE hr , RFID_TAG_HANDLE ht, BYTE *EASFlag) ;			
Instruction	Check EAS enable status			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle, created by ISO15693_connect
	EASFlag	BYTE *	[out]	Return to EAS enable status
Return	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.8.5. NXPICODESLI_GetRandomAndSetPassword 【 Passwork Authorized Command】

Definition	err_t NXPICODESLI_GetRandomAndSetPassword (RFID_READER_HANDLE hr RFID_TAG_HANDLE ht, BYTE pwdNo, DWORD pwd) ;			
Instruction	Icode slx or Icode slx2 tag password authority, the function get automatically random XOR			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	pwdNo	BYTE	[in]	Password type. EAS/AFI is 10h
	pwd	DWORD	[in]	4bytes authentication password
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.8.6. NXPICODESLI_WritePassword 【Verify Password】

Definition	err_t NXPICODESLI_WritePassword(RFID_READER_HANDLE hr RFID_TAG_HANDLE ht, BYTE pwdNo, DWORD pwd) ;			
Instruction	Change password, use Set Password command to authorize old password before deliver this command, change password would be successfully once pass authorization and password block has not been locked.			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	pwdNo	BYTE	[in]	Password identification sign EAS/AFI is 10h .
	pwd	DWORD	[in]	32bit new password
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.8.7. NXPICODESLI_LockPassword 【Lock Password Area】

Definition	err_t NXPICODESLI_LockPassword(RFID_READER_HANDLE hr RFID_TAG_HANDLE ht, BYTE pwdNo) ;			
Instruction	Lock password block, verifying password would be failure once locked. Need to use Set Password command to authorize present password before deliver this command.			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	pwdNo	BYTE	[in]	Password identification sign EAS/AFI is 10h .
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.8.8. NXPICODESLI_PasswordProtect 【Set Password Protection Area】

Definition	err_t NXPICODESLI_PasswordProtect(RFID_READER_HANDLE hr RFID_TAG_HANDLE ht, BYTE bandType);				
Instruction	Enable EAS or AFI password protection, once enable, set password authorization before verify EAS or AFI				
Parameter	Name	Type	Direction	Remark	
	hr	RFID_READER_HANDLE	[in]	Reader driver handle	
	ht	RFID_TAG_HANDLE	[in]	Tag handle	
	bandType	BYTE	[in]	Identification of tag memory area	
				Value	Memory Area
				0	EAS
				1	AFI
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.				
Reference Code					

3.8.9. NXPICODESLI_ProtectPage 【Set Page Protection】

Definition	<code>err_t NXPICODESLI_ProtectPage(RFID_READER_HANDLE hr , RFID_TAG_HANDLE ht,BYTE PPPointer,BYTE protSta) ;</code>			
Instruction	Protect the data blocks. "PPPointer" is used as the demarcation. The low 4-bit data of "proSta" represents the status of the blocks which address are less than the demarcation. The high 4-bit data of "proSta" represents the status of the blocks which address are equal or greater than the demarcation.			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	PPPointer	BYTE	[in]	The value of the demarcation.
	protSta	BYTE	[in]	- Low 4-bit data :the blocks which address are less than the demarcation. - High 4-bit data: the blocks which address are equal or greater than the demarcation. Description of protected status: 0x00:public, unrestricted; 0x01:read/write protected by password; 0x02:write protected by password; 0x03:Read is protected by read password, write is protected by write password; If 64 bit password is enabled, both read and write operations are protected by read password and write password.
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.8.10. NXPICODESLI_LockPageProtection 【Lock Page protection】

Definition	<code>err_t NXPICODESLI_LockPageProtection(RFID_READER_HANDLE hr , RFID_TAG_HANDLE ht,BYTE pageAddr) ;</code>			
Instruction	Lock the protection status. Once locked successfully, it can't recover to unlock status again.			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	pageAddr	BYTE	[in]	The demarcation of the block.
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.8.11. NXPICODESLI_Enable64BitPwd 【Enable 64-bit Password】

Definition	<code>err_t NXPICODESLI_Enable64BitPwd(RFID_READER_HANDLE hr , RFID_TAG_HANDLE ht)</code>			
Instruction	Enable 64-bit password protected.			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle.
	ht	RFID_TAG_HANDLE	[in]	Tag handle.
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.8.12. NXPCODESLI_EnblePrivacy 【Enable Privacy】

Definition	<code>err_t NXPCODESLI_EnblePrivacy(RFID_READER_HANDLE hr , RFID_TAG_HANDLE ht,DWORD pwd) ;</code>			
Instruction	Enable tag privatization. Once the privatization is completed, after the tag is rewritten and reset, it will not respond to any command until the privatization password is authenticated			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle.
	ht	RFID_TAG_HANDLE	[in]	Tag handle.
	pwd	DWORD	[in]	The privacy password.
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.8.13. NXPICODESLI_Destroy 【Destroy Tag】

Definition	<code>err_t RFIDLIB_API NXPICODESLI_Destroy(RFID_READER_HANDLE hr , RFID_TAG_HANDLE ht,DWORD pwd)</code>			
Instruction	Destroy tag, if destroy, tag can not be use again			
Parameter	Name	Type	Dire ction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle.
	ht	RFID_TAG_HANDLE	[in]	Tag handle.
	pwd	DWORD	[in]	The password which use for destroy tag.
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.8.14. NXPICODESLI_ReadSignature 【Read Signature】

Definition	<code>err_t NXPICODESLI_ReadSignature(RFID_READER_HANDLE hr , RFID_TAG_HANDLE ht,BYTE signature[]) ;</code>			
Instruction	Read the signature of the tag.			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle.
	ht	RFID_TAG_HANDLE	[in]	Tag handle.
	signature	BYTE []	[out]	The buffer used for saving the 32 bytes data of signature.
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.8.15. NXPICODESLI_WriteEASID 【Write EAS ID】

Definition	<code>err_t NXPICODESLI_WriteEASID(RFID_READER_HANDLE hr , RFID_TAG_HANDLE ht,WORD EASID)</code>			
Instruction	Write EAS ID.			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle.
	ht	RFID_TAG_HANDLE	[in]	Tag handle.
	EASID	WORD	[IN]	16-bit EAS ID
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.8.16. NXPICODESLI_PresetCounter 【Preset Counter】

Definition	<code>err_t RFIDLIB_API NXPICODESLI_PresetCounter(RFID_READER_HANDLE hr , RFID_TAG_HANDLE ht,WORD initCnt,BOOLEAN enReadPwdProtect)</code>			
Instruction	Preset the counter of the tag.			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle.
	ht	RFID_TAG_HANDLE	[in]	Tag handle.
	initCnt	WORD	[in]	Initial value of counter
	enReadPwdProtect	BOOLEAN	[in]	Whether to enable read password protection
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.8.17. NXPICODESLI_IncrementCounter 【Counter plus 1】

Definition	<code>err_t NXPICODESLI_IncrementCounter(RFID_READER_HANDLE hr , RFID_TAG_HANDLE ht) ;</code>			
Instruction	The counter plus 1.			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle.
	ht	RFID_TAG_HANDLE	[in]	Tag handle.
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.8.18. NXPICODESLI_GetNxpSysInfo 【Get System Information】

Definition	err_t NXPICODESLI_GetNxpSysInfo(RFID_READER_HANDLE hr , RFID_TAG_HANDLE ht,BYTE *PPPointer ,BYTE *PPConditions , BYTE *lockBits,DWORD *featureFlags) ;			
Instruction	Get specific system information of NXP tag.			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle.
	ht	RFID_TAG_HANDLE	[in]	Tag handle.
	PPPointer	BYTE *	[out]	Watershed address of the protection page
	PPConditions	BYTE *	[out]	Status of page protection
	lockBits	BYTE *	[out]	AFI,DSFID ,EAS ,Lock of page protection status
	featureFlags	DWORD *	[out]	NXP icode sli special function flag.
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.9. ST M24LR Self-definition Command

3.9.1. STM24LR_WriteSectorPassword 【Set Password】

Definition	err_t RFIDLIB_API STM24LR_WriteSectorPassword(RFID_READER_HANDLE hr, RFID_TAG_HANDLE ht, BYTE pwdnum, DWORD new_pwd)			
Instruction	Update RF password			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	pwdnum	BYTE	[in]	RF password area number. 1-3
	new_pwd	DWORD	[in]	Password. DWORD conversion example : 4 byte password 0x01,0x02,0x03,0x04 Convert to DWORDtype 0x04030201.
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.9.2. STM24LR_PresentSectorPassword 【Authorized Password】

Definition	err_t RFIDLIB_API STM24LR_PresentSectorPassword(RFID_READER_HANDLE hr, RFID_TAG_HANDLE ht, BYTE pwdnum, DWORD pwd)			
Instruction	Authentication sector password. If the password protection of a sector is activated, the API must be used for authentication before changing the data of the sector data block, otherwise the write fails			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	pwdnum	BYTE	[in]	RF password area number. 1-3
	pwd	DWORD	[in]	32bit password。 DWORD conversion example : 4 byte password 0x01,0x02,0x03,0x04 Convert to DWORDtype 0x04030201.
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.9.3. STM24LR_ReadCFG 【Read Configuration BYTE】

Definition	err_t STM24LR_ReadCFG (RFID_READER_HANDLE hr , RFID_TAG_HANDLE ht,BYTE *cfgby)			
Instruction	Read 8-bit configuration byte area of system storage area			
Parameter	Name	Type	Direction	
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	cfgby	BYTE*	[in]	Configure the value of bytes
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference				

3.9.4. STM24LR_WriteEHCfg

Definition	err_t STM24LR_WriteEHCfg (RFID_READER_HANDLE hr , RFID_TAG_HANDLE ht, BYTE EnergyHarvesting, BYTE EHMode) ;				
Instruction	Set up the "energy collection" function in the configuration storage area.				
Parameter	Name	Type	Direction	Remark	
	hr	RFID_READER_HANDLE	[in]	The reader handle	
	ht	RFID_TAG_HANDLE	[in]	Tag handle	
	EnergyHarvesting	BYTE	[in]	Power level, 0-3.	
	EHMode	BYTE	[in]	Mode	
				Value	description
				0	After startup, the "Energy Collection" function will automatically open.
				1	After startup, the "Energy Collection" function will can not automatically
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.				
Reference					

3.9.5. STM24LR_SetRstEHEn

Definition	err_t STM24LR_SetRstEHen (RFID_READER_HANDLE hr , RFID_TAG_HANDLE ht, BYTE enable) ;				
Instruction	Enable of diable control register “energy collection” function				
Parameter	Name	Type	Direction	Remark	
	hr	RFID_READER_HAN DLE	[in]	Reader handle	
	ht	RFID_TAG_HANDLE	[in]	Tag handle	
	enable	BYTE	[in]	Enable/Disable	
				Value	description
				0	Disable
				1	Enable
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.				
Reference					

3.9.6. STM24LR_WriteDOCfg

Definition	err_t STM24LR16E_ReadCtrlReg(RFID_READER_HANDLE hr , RFID_TAG_HANDLE ht, BYTE * cfg_do) ;			
Instruction				
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	cfg_do	BYTE	[in]	Register value
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference				

3.10. CIT83128 Self-definition Command

3.10.1. CIT83128_InitMem

Definition	err_t CIT83128_InitMem(RFID_READER_HANDLE hr,RFID_TAG_HANDLE ht, BOOLEAN loseEffect, BYTE blkAddr,BYTE blkData[]) ;			
Instruction	Initialize memory			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	loseEffect	BOOLEAN	[in]	Whether command invalid, 0: command valid 1: command invalid
	blkAddr	BYTE	[in]	Data block Address
	blkData	BYTE[]	[in]	4-byte block data
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.10.2. CIT83128_ActAu

Definition	err_t CIT83128_ActAu(RFID_READER_HANDLE hr , RFID_TAG_HANDLE ht, DWORD rr, BYTE token[]);			
Instruction	This command is used to sponsor unidirectional discrimination to tag chip.			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	rr	DWORD	[in]	32bits random number
	token	BYTE[]	[out]	8bytes discriminate data
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code	No			

3.10.3. CIT83128_GetSecPara

Definition	err_t RFIDLIB_API CIT83128_GetSecPara(RFID_READER_HANDLE hr , RFID_TAG_HANDLE ht, BOOLEAN readSecByte,BYTE secData[])			
Instruction	Get security parameter of chip			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	readSecByte	BYTE	[in]	Whether read security byte. 0: don't read, chip return 4-byte data. 1: read, chip return 5-byte data
	secData	BYTE[]	[out]	4 or 5bytes security data
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.11. ST25DV04K

3.11.1. ST25DV04K_ReadConfiguration

Definition	err_t RFIDLIB_API ST25DV04K_ReadConfiguration (RFID_READER_HANDLE hr, RFID_TAG_HANDLE ht, BYTE Pointer, BYTE *RegisterValue)			
Instruction	Obtain the static configuration register value of the chip			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	Pointer	BYTE	[in]	Register address
	RegisterValue	BYTE*	[out]	Register value
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.11.2. ST25DV04K_WriteConfiguration

Definition	err_t RFIDLIB_API ST25DV04K_WriteConfiguration (RFID_READER_HANDLE hr, RFID_TAG_HANDLE ht, BYTE Pointer, BYTE RegisterValue)			
Instruction	Set the static configuration register value of the chip. Before operating, invoke ST25DVxx_PresentPassword to authentication RF_PWD_0			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	Pointer	BYTE	[in]	Register address
	RegisterValue	BYTE	[in]	The value of register
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.11.3. ST25DV04K_ReadDynamicConfiguration

Definition	err_t RFIDLIB_API ST25DV04K_ReadDynamicConfiguration (RFID_READER_HANDLE hr,RFID_TAG_HANDLE ht, BYTE Pointer, BYTE *RegisterValue)			
Instruction	read the register value of chip			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	Pointer	BYTE	[in]	Register address
	Register Value	BYTE*	[out]	The value of register
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.11.4. ST25DV04K_WriteDynamicConfiguration

Definition	err_t RFIDLIB_API ST25DV04K_WriteDynamicConfiguration (RFID_READER_HANDLE hr, RFID_TAG_HANDLE ht, BYTE Pointer, BYTE RegisterValue)			
Instruction	Set the dynamic register value of chip			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	Pointer	BYTE	[in]	Register address
	RegisterValue	BYTE	[in]	The value of register
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.11.5. ST25DV04K_ManageGPO

Definition	err_t RFIDLIB_API ST25DV04K_ManageGPO (RFID_READER_HANDLE hr, RFID_TAG_HANDLE ht, BYTE gpo)			
Instruction	set the value of GPOVAL			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	gpo	BYTE	[in]	Value of GPOVAL
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.11.6. ST25DV04K_WriteMessage

Definition	err_t RFIDLIB_API ST25DV04K_WriteMessage (RFID_READER_HANDLE hr, RFID_TAG_HANDLE ht, BYTE* msgData, BYTE msgLen)			
Instruction	Write Mailbox data of tag			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	msgData	BYTE*	[in]	Buffer data to written
	msgLen	BYTE	[in]	Data length of buffer
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.11.7. ST25DV04K_ReadMessageLength

Definition	err_t RFIDLIB_API ST25DV04K_ReadMessageLength (RFID_READER_HANDLE hr, RFID_TAG_HANDLE ht, BYTE* msgLength)			
Instruction	Read the length of buffer data for tag Mailbox			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	msgLength	BYTE	[out]	Data length of buffer
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.11.8. ST25DV04K_ReadMessage

Definition	err_t RFIDLIB_API ST25DV04K_ReadMessage (RFID_READER_HANDLE hr, RFID_TAG_HANDLE ht, BYTE Pointer, BYTE ReadCount, BYTE *msgBuf, DWORD *nSize)			
Instruction	Read buffer data of tag Mailbox			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	Pointer	BYTE	[in]	address
	ReadCount	BYTE	[in]	Data length of buffer to read
	msgBuf	BYTE*	[out]	Data buffer
	nSize	DWORD	[in] [out]	[in]: length of msgBuf [out]:Data length of buffer that read success
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.11.9. ST25DV04K_WritePassword

Definition	err_t RFIDLIB_API ST25DV04K_WritePassword (RFID_READER_HANDLE hr, RFID_TAG_HANDLE ht, BYTE passwordNum, BYTE password[])			
Instruction	Set RF configuration password			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	passwordNum	BYTE	[in]	Password type: 00h: RF_PWD_0 01h: RF_PWD_1 02h: RF_PWD_2 03h : RF_PWD_3
	password	BYTE[]	[in]	8 -byte length of password
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.11.10. ST25DV04K_PresentPassword

Definition	err_t RFIDLIB_API ST25DV04K_PresentPassword (RFID_READER_HANDLE hr, RFID_TAG_HANDLE ht, BYTE passwordNum, BYTE password[]/*8 bytes*/)			
Instruction	Authentication RF configuration password			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	passwordNum	BYTE	[in]	Password type: 00h: RF_PWD_0 01h: RF_PWD_1 02h: RF_PWD_2 03h : RF_PWD_3
	password	BYTE[]	[in]	8 -byte length of password
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.12. EM4237SLIX/ EM4237SLIC

3.12.1. EM4237SLI_EnableEAS

Definition	err_t RFIDLIB_API EM4237SLI_EnableEAS (RFID_READER_HANDLE hr, RFID_TAG_HANDLE ht)			
Instruction	Enable EAS			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.12.2. EM4237SLI_DisableEAS

Definition	err_t RFIDLIB_API EM4237SLI_DisableEAS (RFID_READER_HANDLE hr, RFID_TAG_HANDLE ht)			
Instruction	Forbidden EAS			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.12.3. EM4237SLI_LockEAS

Definition	err_t RFIDLIB_API EM4237SLI_LockEAS (RFID_READER_HANDLE hr, RFID_TAG_HANDLE ht)			
Instruction	Lock EAS			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.12.4. EM4237SLI_ActiveEAS

Definition	err_t RFIDLIB_API EM4237SLI_ActiveEAS(RFID_READER_HANDLE hr, RFID_TAG_HANDLE ht, BYTE AddressMode, BYTE EASIdMaskLen, WORD EASID, BYTE EASData[], DWORD*nSize)			
Instruction	Check EAS, when EAS close or unmatched corresponding tag, function will return fail.			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	AddressMode	BYTE	[in]	Address mode: 0: no address 1: match EASID
	EASIdMaskLen	BYTE	[in]	Length of EASID, unit: bit. Effect when AddressMode=1
	EASID	WORD	[in]	EASID value
	EASData	BYTE*	[out]	buffer to save EAS
	nSize	DWORD*	[in] [out]	[in]:The maximum number of buffer that can save EAS [out]: Indicates the actual EAS data size
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.12.5. EM4237SLI_ProtectEAS

Definition	err_t RFIDLIB_API EM4237SLI_ProtectEAS(RFID_READER_HANDLE hr, RFID_TAG_HANDLE ht)			
Instruction	Protected EAS			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.12.6. EM4237SLI_EASCheck

Definition	err_t RFIDLIB_API EM4237SLI_EASCheck(RFID_READER_HANDLE hr, RFID_TAG_HANDLE ht,BYTE *EASFlag)			
Instruction	Check EAS, when EAS close or unmatched corresponding tag, function will return fail			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	EASFlag	BYTE *	[out]	EAS enable status
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.12.7. EM4237SLI_WriteEASID

Definition	err_t RFIDLIB_API EM4237SLI_WriteEASID(RFID_READER_HANDLE hr, RFID_TAG_HANDLE ht, WORD EASID)			
Instruction	Set EASID value			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	EASID	WORD	[in]	EASIDvalue
return(err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.12.8. EM4237SLI_WriteEASCfg

Definition	err_t RFIDLIB_API EM4237SLI_WriteEASCfg(RFID_READER_HANDLE hr, RFID_TAG_HANDLE ht, WORD cfg)			
Instruction	Set EAS configuration information			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	cfg	WORD	[in]	EAS configuration value: 0: 256 bit EAS information (block24~block31) 1: 128 bit EAS information (block28~block31) 2: 64 bit EAS information (block30~block31) 3: 32 bit EAS info (block31)
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.12.9. EM4237SLI_ChangeKey

Definition	err_t RFIDLIB_API EM4237SLI_ChangeKey(RFID_READER_HANDLE hr, RFID_TAG_HANDLE ht, BYTE KeyIdentifier, DWORD Password)			
Instruction	Set password			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	KeyIdentifier	BYTE	[in]	Password index which will be modified: 0: Login password Other: Reserved
	Password	DWORD	[in]	password
return(err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.12.10. EM4237SLI_ProtectMemoryPage

Definition	err_t RFIDLIB_API EM4237SLI_ProtectMemoryPage(RFID_READER_HANDLE hr, RFID_TAG_HANDLE ht, DWORD PageNo, BYTE ProtectStatus)			
Instruction	Page protected			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	PageNo	DWORD	[in]	Page address
	ProtectStatus	BYTE	[in]	Status: 0: both read/write are not protected 1: read protected 2: write protected 3: both read/write are protected
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.12.11. EM4237SLI_Destroy

Definition	err_t RFIDLIB_API EM4237SLI_Destroy(RFID_READER_HANDLE hr, RFID_TAG_HANDLE ht)			
Instruction	Destroy tag			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.12.12. EM4237SLI_EnablePrivacy

Definition	err_t RFIDLIB_API EM4237SLI_EnablePrivacy (RFID_READER_HANDLE hr, RFID_TAG_HANDLE ht)			
Instruction	enable privatization			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.12.13. EM4237SLI_DisablePrivacy

Definition	err_t RFIDLIB_API EM4237SLI_DisablePrivacy(RFID_READER_HANDLE hr, RFID_TAG_HANDLE ht)			
Instruction	Cancel privatization			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.12.14. EM4237SLI_EnableRandomID

Definition	err_t RFIDLIB_API EM4237SLI_EnableRandomID(RFID_READER_HANDLE hr, RFID_TAG_HANDLE ht)			
Instruction	Enable random UID function			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.12.15. EM4237SLI_DisableRandomID

Definition	err_t RFIDLIB_API EM4237SLI_DisableRandomID (RFID_READER_HANDLE hr, RFID_TAG_HANDLE ht)			
Instruction	Cancel random UID function			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.12.16. EM4237SLI_Login

Definition	err_t RFIDLIB_API EM4237SLI_Login(RFID_READER_HANDLE hr, RFID_TAG_HANDLE ht, DWORD password)			
Instruction	Login			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	password	DWORD	[in]	password
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.13. Fudan Microelectronic Self-definition Command

3.13.1. FM13HF_PadIOOnOff

Definition	err_t FM13HF_PadIOOnOff(RFID_READER_HANDLE hr, RFID_TAG_HANDLE ht, BYTE ledCfg1, BYTE ledCfg2);			
Instruction	Used to control the IO pin to output an open drain signal at a certain frequency. Currently, the label model that supports the interface is FM13HS01			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	ledCfg1	BYTE	[in]	Signal output type: 0: High level; 1: Low level; 2: High low switching
	ledCfg2	BYTE	[in]	The frequency of high and low switching, it is only effective when ledCfg1=2. 0: high and low 77ms; 1: high and low 38ms; 2: high and low 154ms; 3: high and low 308ms;
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.13.2. FM11NP04V.Write Reg

Definition	<code>err_t FM11NP04V_WriteReg(RFID_READER_HANDLE hr, RFID_TAG_HANDLE ht, BYTE regAddr, BYTE regDLen, BYTE regDat[]);</code>			
Instruction	Write register			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	regAddr	BYTE	[in]	Register address
	regDLen	BYTE	[in]	The length of the data to be written, up to 4 bytes
	regDat	BYTE[]	[in]	Register data
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.13.3. FM11NP04V.Read Reg

Definition	<code>err_t FM11NP04V_ReadReg(RFID_READER_HANDLE hr, RFID_TAG_HANDLE ht, BYTE regAddr, BYTE regDat[]);</code>			
Instruction	Read register			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	regAddr	BYTE	[in]	Register address
	regDat	BYTE[]	[out]	The register data have been read
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.13.4. FM11NP04V.Read Memory

Definition	<code>err_t FM11NP04V_ReadMemory(RFID_READER_HANDLE hr, RFID_TAG_HANDLE ht, BYTE blkAddr, BYTE blkDat[]);</code>			
Instruction	Read the data block except the user area			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	blkAddr	BYTE	[in]	Data block address
	blkDat	BYTE[]	[out]	Block data have been read
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.13.5. FM11NP04V.Write Memory

Definition	<code>err_t RFIDLIB_API FM11NP04V_WriteMemory(RFID_READER_HANDLE hr, RFID_TAG_HANDLE ht, BYTE blkAddr, BYTE blkDat[]);</code>			
Instruction	Write data blocks except the user area			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	blkAddr	BYTE	[in]	Data block address
	blkDat	BYTE[]	[in]	Block data to be written
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.13.6. FM11NP04V.Password Authenticate

Definition	<code>err_t FM11NP04V_PWDAuth(RFID_READER_HANDLE hr, RFID_TAG_HANDLE ht, DWORD pwd);</code>			
Instruction	32-bit password authentication			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	pwd	DWORD	[in]	password
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.13.7. FM11NT083V.Write FIFO

Definition	<code>err_t FM11NT083V_WriteFIFO(RFID_READER_HANDLE hr, RFID_TAG_HANDLE ht, BYTE FIFODat[], BYTE DLen);</code>			
Instruction	Write FIFO			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	FIFODat	BYTE[]	[in]	FIFO data
	DLen	BYTE	[in]	Data length
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				

3.13.8. FM11NT083V.Read FIFO

Definition	<code>err_t FM11NT083V_ReadFIFO(RFID_READER_HANDLE hr, RFID_TAG_HANDLE ht, BYTE DLen, BYTE Dat[], BYTE *nRead);</code>			
Instruction	Read FIFO			
Parameter	Name	Type	Direction	Remark
	hr	RFID_READER_HANDLE	[in]	Reader driver handle
	ht	RFID_TAG_HANDLE	[in]	Tag handle
	DLen	BYTE	[in]	Data length to read
	Dat	BYTE[]	[out]	Data buffer have been read
	nRead	BYTE *	[out]	Return the length of data read and written to the buffer
Return (err_t)	0: Success Not 0: Failure, please check RFIDLIB Error code table.			
Reference Code				